

Symmetric Key

Sender

Unsecure Channel

Receiver

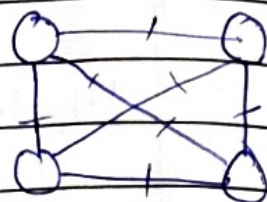
$$C = [K, M]$$

we use
key to encrypt

four four nodes
we require how many
keys?



encryption & decryption
from the same
Key.

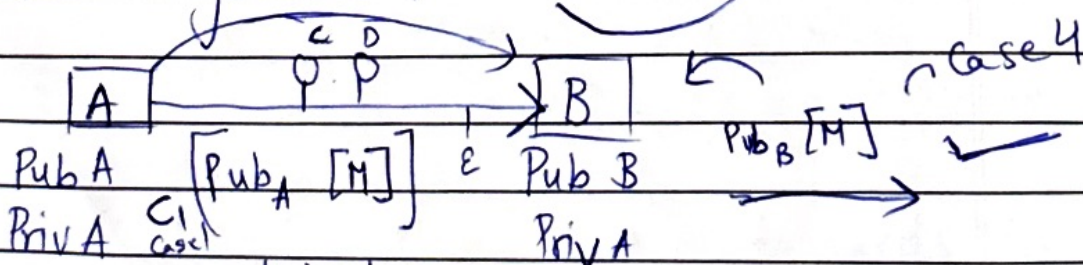


$$\underline{nC_2}$$

$$\begin{aligned} \text{Formula} &= {}^4C_2 \\ &= \frac{4!}{2! \times 2!} = \frac{2 \times 4 \times 3 \times 2 \times 1}{2 \times 2} \\ &= 6 \text{ keys} \end{aligned}$$

Asymmetric Key or "Public Key" (RSA)

uses two keys - Public & Private



encrypt

Case 2

No b.dy
Can open
except A

c_2 Pri B [M] → X

c_3 Pri A [M] → X

Invalid

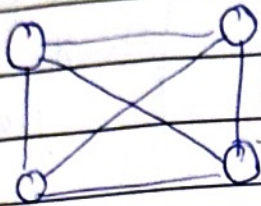
$Pub_B[M] \xrightarrow{P_B} [B]$

RSA - Rivest
Shamir - Adleman

Receiver

always use receiver's Public Key to encrypt the message

will use the private key to open the message



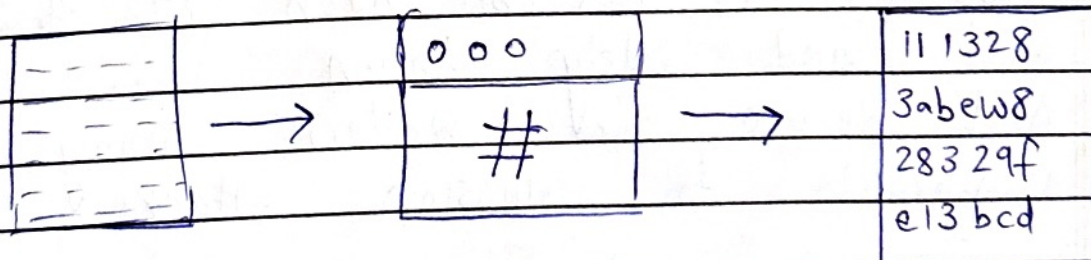
$$\begin{aligned} \text{Formula} &= 2n \\ &= 2 \times 4 \\ &= 8 \end{aligned}$$

Hashing takes input & transforms it into fixed size string

* Use of Hashing

1. Data comparison
2. Integrity of data
3. Password Security

digest

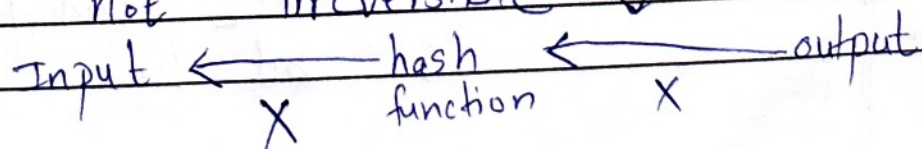


Input File

Hashing
Algorithm
MD5, SHA1
SHA256

Digest /
Hash value

It is not irreversible



Qualities of hash function

1. Fixed size hash value.

If MD5 hash algorithm is used

Input	Hash Value	Hash size
a	00c17b960b92kc399	128 bit
HellothisisNavya	e4f58a805a6effd0f6	<u>128 bit</u>

2. Output is unique for every input.

3. Hash value is irreversible

4. Used to store data in database

MD5 (Message Digest Algorithm 5)

- Produces a 128 bit hash (32 hexadecimal characters)
- fast and widely known
- Not secure for modern security purposes
- vulnerable to collision attacks

eg. MD5

Input: Cybersecurity

MD5: 4d0f7ec3f5f67d4f2d5f8c8f6b5a7d2e

SHA - 256 (Secure Hash Algorithm)

- part of the SHA-2 family
- produces a 256-bit hash

- much more secure than MD5
- Currently considered secure for most applications

Digital Certificates

electronic document used to verify the identity of a website, server, or user on a network.

contains -

- certificate owner's name
- public key
- certificate serial number
- expiry date
- issuing authority (CA)
- digital signature of the CA.

purpose -

- authentication (proves identity)
- secure communication
- prevents impersonation attacks.

eg. When you visit a website with https, your browser checks the website's digital certificate to verify that the site is genuine.

IP Sec (Internet Protocol Security)

- Secures IP communications at the Network layer. (layer 3) of the OSI Model.
- protection for IP packets integrity

features -

- supports two main security protocols -
- * Authentication Header (AH) for authentication and integrity protection.
- * Encapsulating Security Payload for encryption & Confidentiality.

usage -

- used for site to site VPNs where secure tunnels are established b/w 2 or more networks

SSH (Secure Shell)

- operates on (layer 7) of the OSI
- secures remote login
- file transfer over insecure networks
- provides encrypted communication
- accessing remote servers
- uses strong encryption algorithms for data confidentiality.
- Public - Key cryptography for authentication
- Secure channel forwarding for tunneling other protocols

SSL / TLS (Secure Sockets Layer / Transport Layer Security)

- operates on transport layer (layer 4)
- implemented as part of application protocols such as HTTP, SMTP, and IMAP.

- Supports various cryptographic algorithms for encryption (eg. AES, DES), authentication (RSA, ECDSA) and key exchange (Diffie - Hellman). Provides secure communication.

Usage -

- used for securing websites (https), email communications (SMTP, IMAPs), virtual private networks (VPNs) and other secure applications.

Encrypt / Decrypt Commands using OpenSSL

crypter navya.txt

crypter encrypt navya.txt (Encrypts text)

crypter decrypt navya.txt (decrypts text)